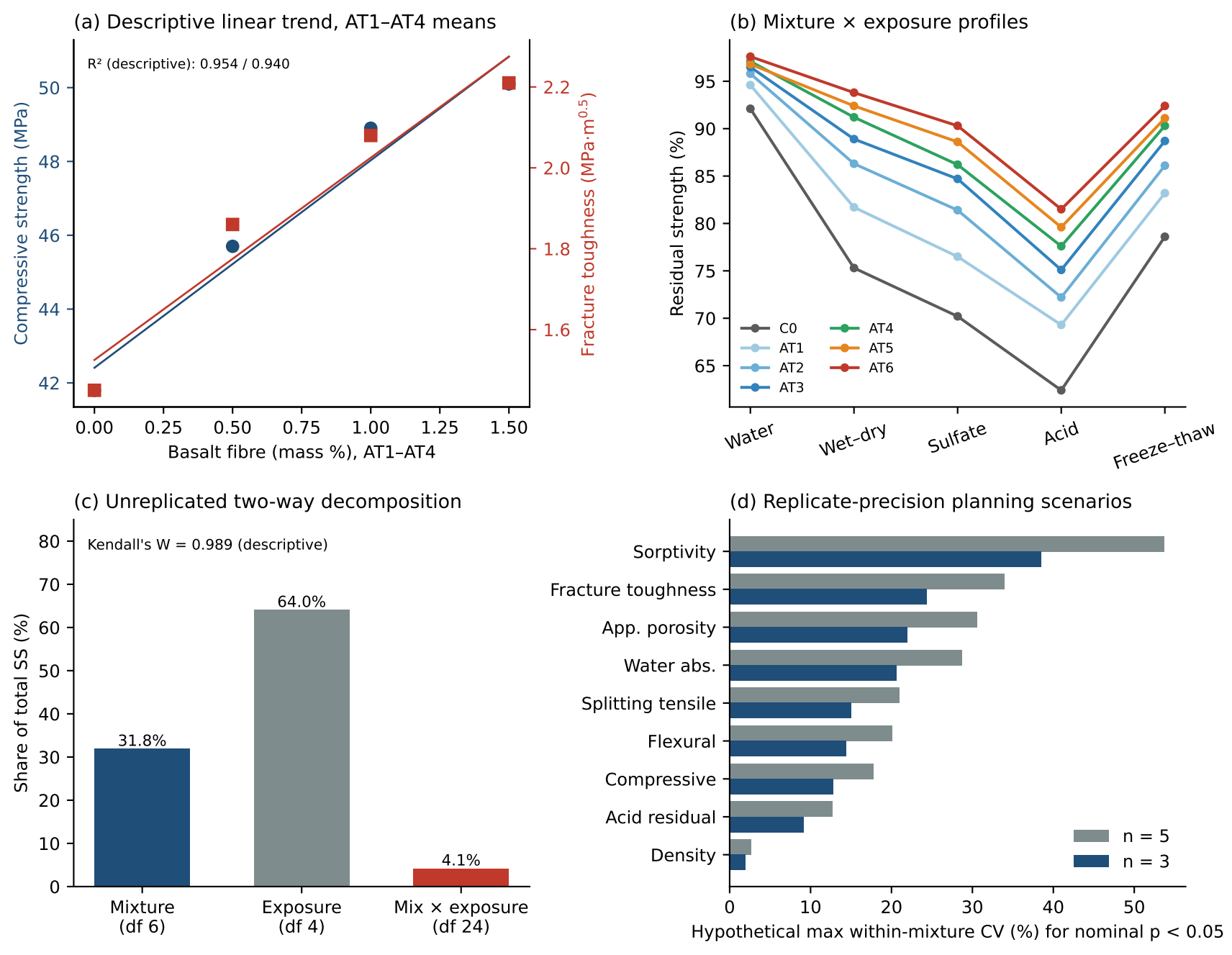
(a) Descriptive linear trend, AT1–AT4 means
R² (descriptive): 0.954 / 0.940
Compressive strength (MPa)
Fracture toughness (MPa·m^0.5)
Basalt fibre (mass %), AT1–AT4
(b) Mixture × exposure profiles
Residual strength (%)
C0
AT1
AT2
AT3
AT4
AT5
AT6
Water
Wet–dry
Sulfate
Acid
Freeze–thaw
(c) Unreplicated two-way decomposition
Kendall's W = 0.989 (descriptive)
Share of total SS (%)
31.8%
64.0%
4.1%
Mixture (df 6)
Exposure (df 4)
Mix × exposure (df 24)
(d) Replicate-precision planning scenarios
Sorptivity
Fracture toughness
App. porosity
Water abs.
Splitting tensile
Flexural
Compressive
Acid residual
Density
n = 5
n = 3
Hypothetical max within-mixture CV (%) for nominal p < 0.05